⇒ Types of ML :

1. Supervised learning :
2. Unsupervised learning.
3. Reinforcement learning.

1. Supervised learning :-

The computer is given labeled data. This means you provide both the input question AND the correct answer (label), the algorithm learns the pattern b/w the two so it can guess the answer for new, unseen inputs.

ex: Student Exam prediction!

Study hours per week (x1)	Attendance Percentage (x2)	Practice Tests Taken (x3)
12	90%	5
3	55%	1
8	80%	4
2	60%	0

How supervised learning works here:

1. The 3. Features (X):
- X_1 = Study Hours.
- X_2 = Attendance %
- X_3 = Practice Tests

2/ The label (Y):
- 1 = Pass, 0 = Fail

3/ Learning the Pattern: The algortham process these rows and spots a rule, such as "If a student studies more than 5 hours AND has above 75% attendance, they usually Pass (1).

4/ Teaching a New Student (unseen) input

Exam outcome (Y-label)
1 (Pass)
0 (Fail)
1 (Pass)
0 (Fail)

You enter a new student's record: [10 hours, 85% attendance, 3 tests].

The model process these 3 numbers and predicts: 1 (Pass)

2, Unsupervised Learning ("Learning On your Own")

Def: The computer is Given unlabeled (data without any correct answer). The algorithm's job is to explore the data and find hidden patterns, clusters, or groupings on its own

ex:-

Dataset Online store ke customers:
Hum e-commerce website 4 customers ko data collect karte hain bina kisi answer/label ke.

Monthly visits (x1)	Items Viewed (x2)	Money spent in $ (x3)
25	120	$ 500
2	50	$ 20
30	150	$ 800
3	8	$ 15

Note: Here we dont have pass/Fail or "Good/bad"
There is no column of answer

So How algorithm works?

1. Numbers comparison (Distance check)
Algorithm compare all numbers with each other.

- customer 1 ([25, 125, 500]) and customer 3 ([30, 150, 800]) ~~ke~~ numbers are larger and they are nearby with each Other!

2/ Groups (clusters) creatinge
it will find pattern by their own,
Algorithm creates 2 groups:

- Group A (High Activity): customer 1 and customer 3
- Group B (Low Activity): customer 2 and customer 4

3/ (Business Decision):
Algorithm just creates groups, then humans assign the name to that Group!

ex:-

- Group A = "vip shoopers"
- Group B = casual Browsers

Group	Label
?	
?	
?	
?	
?	

4. Reinforcement learning ("Learning through Rewards and Penalties")

Defi:- There is no dataset before hand, instead, an AI "Agent" learns by interacting with an environment through trial and error.

- if it makes a good move → it gets Reward (+pts)
- if it makes a bad move → it gets an Penalty (-pts)

Over millions of attempts, the AI learn the best strategy to maximize its total reward.

ex:- Training a puppy. when the puppy sits on command, you give it a treat (Reward) when it bites the sofa, you say "No"! (Penalty)

Eventually, the puppy learns how to behave to get the maximum number of treats.

No static data set (Trial and Error)

3) Semi-Supervised Learning ("Learning with a little help")

Def: A hybrid of supervised and unsupervised learning. The computer is given a small set of labeled data and large amount of unlabeled data.

Labeling data manually takes huge time and money. Semi-supervised learning uses the small labeled set to get a baseline understanding, then uses the unlabeled data to scale of its learning.

ex: A teacher solves 2 practice problems on the whiteboard and then hands the student 100 practice problems to figure out using the same basic principles